



KRISHNENDU GHOSH

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PROFESSIONAL SUMMARY

Computer Science undergraduate specializing in AI/ML, with hands-on experience building full-stack AI applications using Python, FastAPI, React.js, and Flutter. Skilled in computer vision, transfer learning, and Explainable AI (Grad-CAM, SHAP), with research experience and a track record of shipping deployed AI platforms for agriculture, healthcare, and NLP.

EDUCATION

Bachelor of Technology in Computer Science (AI & ML) — Supreme Knowledge Foundation Group of Institutions, Hooghly, WB (MAKAUT)

Expected Graduation: 2027 | Current CGPA: 7.60 (Upto 6th Semester)

Higher Secondary (PCM & Computer Science), WBCHSE — Uttarpara Amarendra Vidyapith, Hooghly, WB | 2023 | 61.6%

Secondary Education, WBBSE — Uttarpara Amarendra Vidyapith, Hooghly, WB | 2021 | 79.71%

INTERNSHIP / RESEARCH EXPERIENCE

Research Intern — IEEE SMC, Kalyani Government Engineering College (KSEC)

May 2026 – Aug 2026 | Kalyani, Nadia, West Bengal

- Developed an AI-based Credit Scoring System using machine learning, covering data preprocessing, EDA, model training, and evaluation.
- Worked with Python, Scikit-learn, Pandas, NumPy, FastAPI, and Git; collaborated on integrating the ML model with the application frontend.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, Dart, SQL

AI / ML: Machine Learning, Deep Learning, Computer Vision, Transfer Learning, CNN, RNN, ANN, LSTM, Explainable AI (Grad-CAM, SHAP), LangChain, GPT-4/3.5, Regression, Classification, Clustering, Feature Engineering, Data Preprocessing, Model Evaluation, Ensemble Learning, Imbalanced Classification, Data Mining, NLP

Web & Mobile: React.js, Flutter, HTML, CSS, REST APIs, FastAPI, Streamlit, Android Studio

Databases & Cloud: Firebase Firestore/Auth, MySQL, PostgreSQL, SQLite

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib, Seaborn, XGBoost, MATLAB

Tools & Platforms: Git, GitHub, Google Colab, Jupyter Notebook, VS Code, Google Earth Engine, Linux, Render, Vercel, Netlify

Core CS: Data Structures & Algorithms, Operating Systems, DBMS

PROJECTS

AgriVision — AI-Driven Precision Agriculture Monitoring Platform (Android App) Sept 2025

Tech: Python, FastAPI, Flutter, Firebase, MATLAB, CNN, LSTM, Google Earth Engine, OpenWeather API

- Built a precision agriculture system leveraging hyperspectral imaging, environmental sensors, and weather APIs, with CNN/LSTM models for crop health, vegetation stress, and pest risk prediction.
- Developed a FastAPI backend and Flutter mobile dashboard integrated with Firebase to deliver real-time farm insights using spectral data and soil/weather inputs.

Clean Route — AI-Enabled Smart Waste Management and Route Optimization System (Android App) Nov 2025

Tech: Flutter, Firebase, Python, REST API

- Developed a cross-platform Flutter application with Firebase integration for real-time waste collection monitoring, including authentication and database management.
- Implemented Python-based backend APIs for route optimization and efficient waste collection planning.

TextSage AI — Intelligent Text Summarization Tool (Web App) Feb 2026 | Live: textsage.krishnendughosh.in

Tech: React.js, Tailwind CSS, Python, Node.js, FastAPI, REST API

- Developed an AI-powered NLP platform for summarization, sentiment analysis, and classification by implementing pipelines for keyword extraction, entity recognition, and semantic analysis.
- Built scalable backend APIs for real-time processing and a responsive React frontend, optimizing preprocessing and inference to support applications like document summarization, review analysis, and research assistance.

OncoScan XAI — Explainable Multi-Modal Transfer Learning Framework for AI-Assisted Breast Cancer Detection In Progress

Tech: Python, PyTorch, Radiomics, DenseNet121, Scikit-learn, FastAPI, React.js, Firebase, Grad-CAM, SHAP, REST API

- Developing an AI-assisted breast cancer detection system using multi-modal MRI and mammography data, implementing transfer learning (DenseNet121) with feature fusion to improve classification performance.
- Integrating Explainable AI techniques (Grad-CAM, SHAP) for interpretability, and building an end-to-end pipeline including preprocessing, FastAPI-based inference APIs, React frontend for visualization/reporting, and Firebase for secure data management.

ACHIEVEMENTS & CERTIFICATIONS

- Smart India Hackathon (SIH) 2025** — Qualified in Internal Round (Sep 2025)
- GSSoC 2024** — Open Source Contributor, Ranked in Top 200 (May 2024)

SOFT SKILLS & INTERESTS

Soft Skills: Problem Solving, Analytical Thinking, Research, Team Collaboration, Adaptability, Time Management

Languages: Bengali, English, Hindi

Disclaimer

I solemnly declare that all the above information is correct to the best of my knowledge and belief.

Signature: Krishnendu Ghosh

Date: 18/08/2026